

geared motor, which works as the dispensing unit. The food container is placed above the aquarium tank and contains sufficient food for the fish. A hole/slot is made at the bottom of the container. A suitable bottom lid for the container is attached to the shaft of the motor. A suitable slot is made in the lid, too.

Both the slots (container and lid) are of the same size. These slots are off-centred. The dispenser unit, which includes motor shaft, lid, container and slots, should be properly aligned. The proposed food dispenser assembly unit is shown in Fig. 3. When the lid rotates along the motor's shaft, the slot gets aligned with the slot of the container. As soon as the two slots are aligned, food reserved in the container falls into the aquarium tank due to self-weight.

The sequence of operations to activate the dispensing unit is shown in the block diagram. Two cellphones are needed in this project. The receiver cellphone must be set on auto-answer-

ing mode, so that when the caller calls, it should automatically receive the call.

After a call is established, a pre-defined sequence of numbers on the keypad are to be pressed on the caller's phone. This sequence is defined in Arduino program. IC1 decoder receives key-pressed DTMF signals from the caller's cellphone and decodes it. These decoded signals are obtained on pins 11 through 14 of IC1. These input signals are processed as per Arduino program and the mathematical equations of calculation of decibel and binary values. Binary value is calculated in the program from input signals using the arithmetic equation given below.

$$\text{Binary value} = 1000 \times v_4 + 100 \times v_3 + 10 \times v_2 + v_1$$

where, v_1 - v_4 are outputs from MT8870 DTMF IC.

Here, keys 2 and 5 on the caller's cellphone are used for starting and stopping the DC geared motor, respectively. The program can be modified as

per requirement. Table II shows the binary value for each key.

Software

Arduino program (Aquarium_food_disp.ino) was used for this prototype. It must be uploaded to Arduino

Uno. For this, Arduino IDE, which is an open source software, is needed. In the program, Arduino digital pins 2 through 5 are defined as input.

Main functions of Arduino code are explained below.

pinMode(). This configures the specified pin to behave either as input or output. (See the description of digital pins in Arduino manual for details on the functionality of pins.)

digitalWrite(). If the pin has been configured as output with pinMode(), its voltage will be set to the corresponding value, which is 5V (or 3.3V on 3.3V boards) for high, and

TABLE II
BINARY VALUE FOR EACH KEY

Q4	Q3	Q2	Q1	Key
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	0
1	0	1	1	*
1	1	0	0	#
1	1	0	1	A
1	1	1	0	B
1	1	1	1	C
0	0	0	0	D

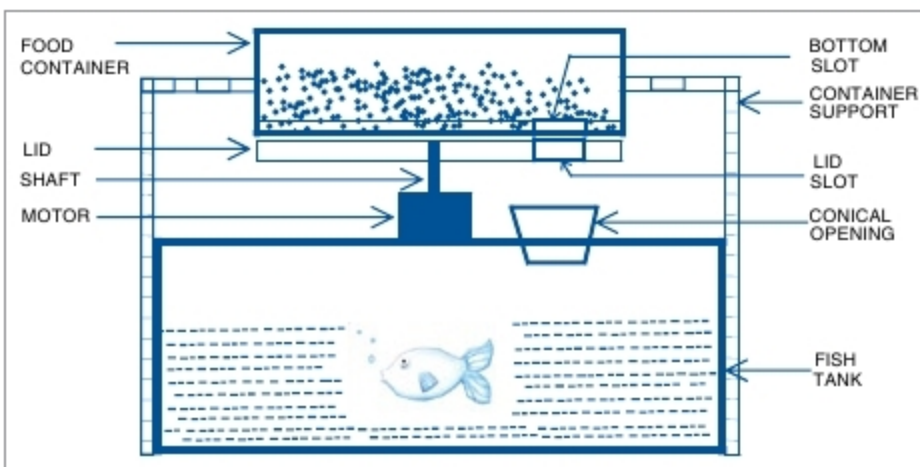


Fig. 3: Proposed food dispenser assembly unit

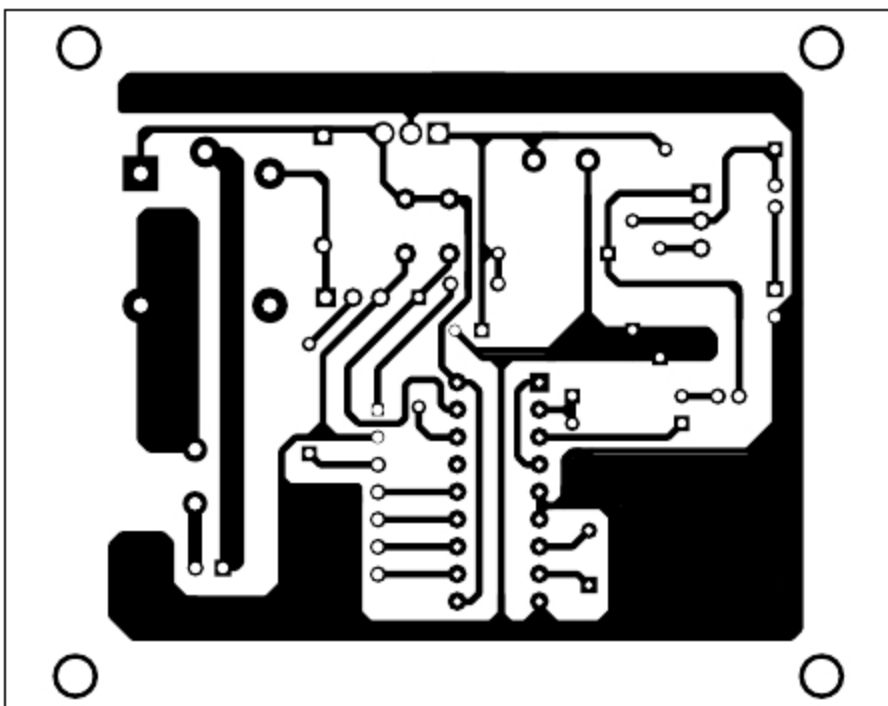


Fig. 4: Actual-size PCB layout of automatic food dispenser

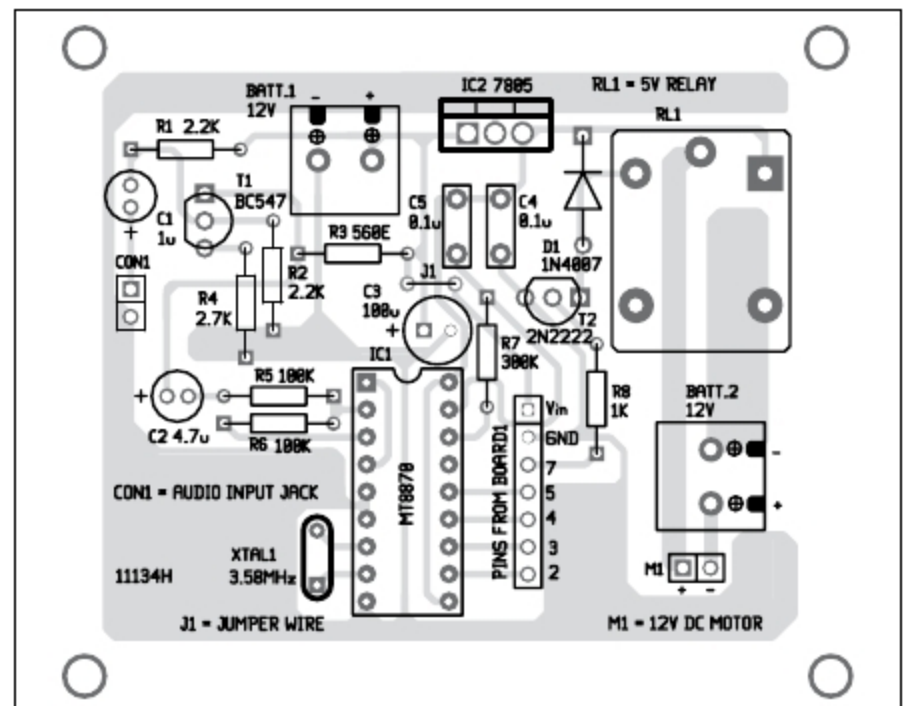


Fig. 5: Components layout for the PCB